THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapters 1–4: **The Wonderful World of Science, Diversity in the Living World, Mindful Eating: A Path to a Healthy Body, Exploring Magnets** | Class 6 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(d)** curiosity	A curious mind keeps asking why and how.
2	**(c)** biodiversity	Biodiversity means diversity of life.
3	**(d)** carbohydrates	Carbohydrates give quick energy.
4	**(a)** magnetic	Magnetic materials include iron, nickel and cobalt.
5	**(a)** a systematic way of understanding nature	Science studies nature through observation and experiment.
6	**(d)** parallel venation	Monocot leaves have parallel veins.
7	**(c)** calcium	Calcium builds bones.
8	**(d)** small magnet	It is a tiny magnet.
9	**(a)** a mobile phone	Technology is the practical use of science.
10	**(b)** herbivores	Cows are herbivores.
11	**(a)** carbohydrates	They contain starch.
12	**(a)** to keep them from losing magnetism	Keepers preserve magnetism.
13	**(b)** observation	Observation is the first step of enquiry.
14	**(a)** habitat	A habitat provides food and shelter.
15	**(a)** proteins	Proteins build and repair tissues.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Repetition checks that the result is reliable and not due to chance (1); It reduces the effect of errors (1).	1+1
17	The variety of living organisms on Earth (1); It includes plants, animals and microorganisms (1).	1+1
18	Substances in food needed by the body (1); Examples: proteins and vitamins (1).	1+1
19	Magnetic: iron, nickel, cobalt (1); Non-magnetic: wood, plastic, glass (1).	1+1
20	Science: understanding how nature works (1); Technology: using that knowledge to make useful things (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Observe and ask a question (1); Make a hypothesis and test it by experiment (1); Record results and draw a conclusion (1).	1+1+1
22	Tomato: herb (1); Rose: shrub (1); Banyan: tree (1).	1+1+1
23	Rice: carbohydrate for energy (1); Pulses: protein for growth (1); Butter: fat for energy and warmth (1).	1+1+1
24	Move a magnet over the mixture (1); Iron nails stick to the magnet (1); Wooden pieces remain (1).	1+1+1
	OR: Observation: what we see, e.g. the ground is wet (1); Inference: what we conclude, e.g. it has rained (1); Inference should be supported by evidence (1).	1+1+1
25	Goat: herbivore (1); Tiger: carnivore (1); Bear: omnivore (1).	1+1+1
	OR: No single food has all nutrients (1); Each nutrient has a different job (1); Variety makes a balanced diet (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Observation: dry seeds do not sprout (1); Hypothesis: seeds need water (1); Experiment: one dish with water, one dry, same seeds and place (1); Result: only wet seeds sprout (1); Conclusion: seeds need water for germination (1).	1+1+1+1+1
	OR: Herbs: soft green stem, e.g. spinach (1); Shrubs: hard branching near the base, e.g. hibiscus (1); Trees: thick trunk, e.g. mango (1); Climbers: need support, e.g. pea (1); Creepers: spread on ground, e.g. pumpkin (1).	1+1+1+1+1

27	Carbohydrates: energy, rice (1); Proteins: growth, pulses (1); Fats: energy, butter (1); Vitamins: protection, fruits (1); Minerals: bones and blood, milk and spinach (1).	1+1+1+1+1
	OR: Collect various objects (1); Bring a magnet near each (1); Magnetic: iron nail, steel pin (1); Non-magnetic: wood, plastic, glass (1); Record in a table (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapters 1–4: **The Wonderful World of Science, Diversity in the Living World, Mindful Eating: A Path to a Healthy Body, Exploring Magnets** | Class 6 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(b)** iron nail	Iron is magnetic.
2	**(b)** a mango tree	Plants grow, respire and reproduce.
3	**(c)** tap root system	Mustard and carrot have tap roots.
4	**(c)** calcium	Milk contains calcium and protein.
5	**(a)** north-south	It aligns with Earth's field.
6	**(c)** technology	Technology applies science.
7	**(a)** carnivores	Lions are carnivores.
8	**(a)** sunlight	Skin makes vitamin D in sunlight.
9	**(d)** mobile phones and credit cards	They may damage magnetic data.
10	**(d)** hypothesis	A hypothesis is a testable explanation.
11	**(a)** herbs	Herbs have tender stems.
12	**(b)** pulses	Pulses and dals are protein-rich.
13	**(b)** glass marble	Glass is not attracted.
14	**(a)** make sure the result is reliable	Repetition checks reliability.
15	**(d)** fibrous root system	Many similar thin roots arise from the base.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Tap root: carrot, mustard (1); Fibrous: wheat, grass (1).	1+1
17	Diet with all nutrients in proper amounts (1); It maintains health (1).	1+1
18	It is influenced by the Earth's magnetism (1); The Earth behaves like a large magnet (1).	1+1
19	Mobile phone or refrigerator (1); Electric fan or pressure cooker (1).	1+1
20	Long legs and padded feet for sand (1); It can go long without water (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Keep in pairs with keepers (1); Avoid heating and hammering (1); Keep away from electronic gadgets (1).	1+1+1
22	Observation: watching things as they are (1); Experiment: changing a condition to see the effect (1); An experiment is done to test a hypothesis (1).	1+1+1
23	They make food by photosynthesis (1); Leaves capture sunlight (1); Animals depend on this food (1).	1+1+1
24	Iron: spinach, jaggery (1); Iron: dates (1); Iodine: iodised salt (1).	1+1+1
	OR: Roll the magnet in iron filings (1); Filings cling mostly at the ends (1); The ends are the poles (1).	1+1+1
25	Hypothesis: plants grow better in sunlight (1); Keep two identical plants, one in light and one in dark (1); Give same water and soil to both (1).	1+1+1
	OR: Tap root has one main root with branches (1); Fibrous roots are many thin roots of similar size (1); Examples: carrot and wheat (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Physics: matter, energy and forces (1); Chemistry: substances and changes (1); Biology: living things (1); Astronomy: space objects (1); Geology: Earth and rocks (1).	1+1+1+1+1
	OR: Blade or lamina is the broad part (1); Petiole is the stalk (1); Midrib is the main vein (1); Veins carry water and food (1); Pattern of veins is venation (1).	1+1+1+1+1

27	Eat without distraction (1); Chew well (1); Choose healthy food (1); Do not waste (1); Eat at fixed times (1).	1+1+1+1+1
	OR: Magnetised needle on a pivot (1); Dial with directions (1); Hold it flat (1); Needle points north (1); Rotate the dial to match N (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 3

Chapters 1–4: **The Wonderful World of Science, Diversity in the Living World, Mindful Eating: A Path to a Healthy Body, Exploring Magnets** | Class 6 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(b)** dietary fibre	It is the undigested plant material.
2	**(b)** horseshoe magnet	U shaped.
3	**(c)** vaccines that prevent diseases	Vaccines were developed by science.
4	**(c)** omnivores	Humans and crows are omnivores.
5	**(d)** obesity	It has excess fat and sugar.
6	**(c)** sailors	They find directions at sea.
7	**(b)** recording and analysing results	Results are recorded and studied before a conclusion.
8	**(a)** shrubs	Rose and hibiscus are shrubs.
9	**(d)** fats	Fats give about twice the energy of carbohydrates.
10	**(a)** poles	Poles are at the ends.
11	**(b)** Does a seed sprout faster in warm water?	Only measurable questions can be tested.
12	**(a)** leaf	Leaves carry out photosynthesis.
13	**(b)** roughage	Fibre helps bowel movement.
14	**(a)** bar	Bar magnets are common.
15	**(a)** botanist	Botany is the study of plants.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	It transports nutrients (1); It removes wastes and keeps the body cool (1).	1+1
17	They keep the magnetism from fading (1); They complete the magnetic pairs (1).	1+1
18	An intelligent guess that can be tested (1); Example: plants kept in sunlight grow taller than plants kept in the dark (1).	1+1
19	Herbs have soft green stems and are small (1); Shrubs have hard woody stems branching near the base (1).	1+1
20	They provide energy (1); They are the main fuel for activities (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Scurvy: vitamin C (1); Rickets: vitamin D (1); Anaemia: iron (1).	1+1+1
22	Bring the magnet near the compass (1); Note which end attracts the north of the needle (1); That end is the south pole of the magnet (1).	1+1+1
23	Plants need water to live (1); He made an observation (1); Conclusion: water is necessary for plant growth (1).	1+1+1
24	Herb: spinach (1); Shrub: hibiscus (1); Tree: neem (1).	1+1+1
	OR: Eat slowly (1); Chew food well (1); Do not waste food (1).	1+1+1
25	Magnetic: attracted, e.g. iron (1); Non-magnetic: not attracted, e.g. plastic (1); Test with a magnet (1).	1+1+1
	OR: It magnifies very small objects (1); Cells or tiny organisms can be seen (1); They are not visible to the naked eye (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Science is the study of nature through observation and tests (1); Technology uses scientific knowledge to make useful things (1); Science: laws of motion, properties of light (1); Technology: bicycle, telescope (1); Both depend on each other (1).	1+1+1+1+1

	OR: Tap root: one main root, e.g. carrot (1); Fibrous: many similar roots, e.g. wheat (1); Reticulate venation: net-like, e.g. peepal (1); Parallel venation: veins parallel, e.g. grass (1); Roots of grass are fibrous, matching parallel veins (1).	1+1+1+1+1
27	Scurvy: bleeding gums (1); Rickets: bent legs (1); Night blindness: poor dim-light vision (1); Anaemia: tiredness (1); Goitre: swelling in the neck (1).	1+1+1+1+1
	OR: Earth behaves like a bar magnet (1); A freely suspended magnet aligns north-south (1); Compass needle is a magnet (1); It settles in the north-south direction (1); Directions are found from this (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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